KSB MIL CONTROLS LIMITED

USER REQUIREMENTS SPECIFICATION | URS-CV-AUTO-001 | Rev 0

ANNEXURE D

Tag wise specification_Sample Resultant_URS

(File Format: Excel)

Role	OUTPUT — Sample Result File
Applicable Parts	Parts 2, 4 and 5
Description	Sample resultant of tag-wise data extraction — process, design, fluid, and service related information, plus major parameters such as body material and rating. Authoritative definition of the Part 2 extraction output.

KSB MIL Controls Limited — Control Valve Datasheet Parsing Tool | Field Mapping Register

Reference Enquiry DS: 2047-1803-EIN-DAS-0017 (Tags: 1803-FV-10401 | 1803-FV-1104 | 1803-FV-1403) | Reference KSB MIL DS: MIL-FM-07-040 (Solutioneer v2 Output)

Serial Number	Sv2 Field Name	Enquiry Field Name (Customer Spec)	Enquiry Field Value (Customer Spec)	Units	Mapping Category	Field Requirement	Logic	Value for Sv2 Field	Service Dependency	Sector Dependency	Similar Field Terminologies	Similar Field Values	SV2 Spec Row Number	Sv2 Spec Field Name	Notes and Validations	Additional Information
1	**Tag No**	Tag No. (Row 1)	1803-FV-10401 | 1803-FV-1104 | 1803-FV-1403	—	Direct	Mandatory	Copy tag number verbatim from top of Enquiry DS (Row 1, left column)	1803-FV-10401 | 1803-FV-1104 | 1803-FV-1403	No	No	Tag Number; Instrument Tag; Tag ID; FV Tag	FV-XXXX; CV-XXXX; HCV-XXXX	Header	Tag No	Tag number must match the P&ID reference in Row 1. Format: Plant-Instrument-Number.	
2	**Service**	Service (Row 2)	SUSPECT CONDT PUMPS(P-606A/B) MCF | LP STEAM TO E-101 PREHEATER | RCSP HOPPER R-104 AERATION PLANT AIR	—	Direct	Mandatory	Copy service description verbatim from Row 2 of Enquiry DS	SUSPECT CONDT PUMPS(P-606A/B) MCF | LP STEAM TO E-101 PREHEATER | RCSP HOPPER R-104 AERATION PLANT AIR	No	No	Service Description; Valve Duty; Application	—	Row 1	Service	May include equipment tag number in parentheses. Carry forward exactly as written.	
3	**Fluid Name**	Process Fluid (Row 14)	SUSPECT LP CONDENSATE | LP Steam | PLANT AIR	—	Direct	Mandatory	Copy fluid name from Row 14 of Enquiry DS	SUSPECT LP CONDENSATE | Steam | Air	Yes	No	Process Fluid; Fluid; Medium; Service Fluid	Water; Condensate; Steam; LP Steam; Plant Air; Nitrogen; Propylene	Row 2	Fluid Name	Fluid name determines the sizing equation set (liquid, steam, or gas). Cross-check with Row 15 (Upstream Condition) for consistency.	
4	**Fluid Phase**	Upstream Condition (Row 15)	Water | Steam | Gas/Vapor	—	Derived	Mandatory	Map Row 15 value to Sv2 dropdown: Water/Liquid/Condensate → Liquid; Steam/LP Steam/HP Steam → Steam; Gas/Vapor/Air/Nitrogen → Gas	Liquid | Steam | Gas	Yes	No	State; Phase; Fluid State; Condition	Liquid; Vapour; Gas; Steam; Two-Phase	Row 3	Fluid Phase	This selection determines the entire sizing equation path in Solutioneer. An incorrect phase will give a completely wrong Cv result.	
5	**Inlet Pipe Size/Sch**	Line Size and Schedule — Inlet (Row 11)	3 in STD | 12 in XS | 6 in STD	in / Schedule	Direct	Mandatory	Extract inlet size and schedule from Row 11 Inlet column. Parse leading number as nominal size; schedule follows (STD, XS, 40S etc.)	3 in/STD | 12 in/XS | 6 in/STD	No	No	Inlet Line Size; Upstream Line Size; Inlet Pipe Class; NPS	1 in; 2 in; 3 in; 4 in; 6 in; 8 in; 12 in; STD; XS; SCH 40; SCH 80	Row 4	Inlet Pipe Size/Sch	Line size also embedded in Row 3 Line Number tag (e.g. '3"-CL-1803...' → 3 in). Use Row 11 as primary source.	
6	**Outlet Pipe Size/Sch**	Line Size and Schedule — Outlet (Row 11)	3 in STD | 12 in XS | 6 in STD	in / Schedule	Direct	Mandatory	Extract outlet size and schedule from Row 11 Outlet column. Usually same as inlet unless expander is fitted.	3 in/STD | 12 in/XS | 6 in/STD	No	No	Outlet Line Size; Downstream Pipe Size	Same as inlet in most cases	Row 5	Outlet Pipe Size/Sch	If Outlet size differs from Inlet size, an expander/reducer is required. Flag for engineer review.	
7	**Design Pressure (MAX)**	Design Pressure / Temperature (Row 27)	12 kgf/cm^2-g | 8 kgf/cm^2-g | 10 kgf/cm^2-g	kgf/cm^2-g	Direct	Mandatory	Extract maximum design pressure from Row 27 of Enquiry DS. This is the higher of the two values stated.	12 kgf/cm^2-g | 8 kgf/cm^2-g | 10 kgf/cm^2-g	No	No	Design Pressure; Max Design Pressure; DP MAX; MAWP	—	Row 6	Design Temp(MAX/MIN)	Design pressure determines the ASME pressure class (150#, 300#, 600# etc.) per ASME B16.34 P-T tables.	
8	**Design Pressure (MIN)**	Design Pressure / Temperature (Row 27)	FV kgf/cm^2-g | FV kgf/cm^2-g | — (not stated)	kgf/cm^2-g	Direct	Mandatory	Extract minimum design pressure from Row 27. FV = Full Vacuum = -1.033 kgf/cm^2-a (or 0 absolute). Carry as 'FV' text if shown.	FV | FV | —	No	No	Min Design Pressure; Full Vacuum; FV; MDMP	FV; Full Vacuum; -1 barg; Vacuum	Row 7	Design Pressure(MAX/MIN)	FV = Full Vacuum as design condition. See DS Page 2 process notes. Confirm with engineer before populating Sv2.	
9	**Design Temp (MAX)**	Design Pressure / Temperature (Row 27)	150 °C | 220 °C | 65 °C	°C	Direct	Mandatory	Extract maximum design temperature from Row 27 of Enquiry DS.	150 °C | 220 °C | 65 °C	No	No	Design Temperature; Max Temp; MADT; MDMT (max)	—	Row 7	Design Temp(MAX/MIN)	Determines material grade selection (e.g. need for low-temp grades below -29°C, or high-temp grades above 230°C).	
10	**Design Temp (MIN) / MDMT**	Ambient Temperature (Row 5) / Process Notes Page 2	4.4 °C (MDMT from process notes all 3 tags)	°C	Derived	Mandatory	MDMT (Minimum Design Metal Temperature) stated in DS Page 2 Process Notes. Not the same as ambient temperature (Row 5). Extract from notes text.	4.4 °C | 4.4 °C | 4.4 °C	No	No	MDMT; Min Design Metal Temperature; Low Temp; Cold Design Temp	—	Row 7	Design Temp(MAX/MIN)	MDMT drives low-temperature material requirements (impact testing). Located in Page 2 Process Notes, not Row 5.	
11	**Flow Rate (MAX)**	Flow Rate @ Max Flow (Row 18)	10.5 | 6405 | 2640	Am^3/h | kg/h | Nm^3/h	Direct	Mandatory	Extract flow rate value from the '@ Max Flow' column of Row 18. Preserve units exactly as stated.	10.5 m^3/h | 6405 kg/h | 2640 Nm^3/h	Yes	No	Maximum Flow Rate; Flow Max; Qmax; Mass Flow Max	m^3/h; Am^3/h; Nm^3/h; kg/h; kg/s; lb/h	Row 11	Flow rate	Units vary between liquid (Am^3/h or kg/h) and gas (Nm^3/h or kg/h). Check units column in Row 17 header row.	
12	**Flow Rate (NOR)**	Flow Rate @ Norm Flow (Row 18)	9.5 | 5823 | 2400	Am^3/h | kg/h | Nm^3/h	Direct	Mandatory	Extract flow rate value from the '@ Norm. Flow' column of Row 18.	9.5 m^3/h | 5823 kg/h | 2400 Nm^3/h	Yes	No	Normal Flow Rate; Nominal Flow; Design Flow; Qnorm	Same as Max units	Row 11	Flow rate	Normal (design) flow is the primary sizing case. Cv at Normal flow determines valve % travel target.	
13	**Flow Rate (MIN)**	Flow Rate @ Min Flow (Row 18)	4.8 | 2912 | 2920	Am^3/h | kg/h | Nm^3/h	Direct	Mandatory	Extract flow rate value from the '@ Min. Flow' column of Row 18.	4.8 m^3/h | 2912 kg/h | 2920 Nm^3/h	Yes	No	Minimum Flow Rate; Qmin; Low Flow	Same as Max units	Row 11	Flow rate	Minimum flow case checks that valve does not throttle too close to closed position (< 10% travel).	
14	**Inlet Pressure (MAX)**	Inlet Pressure @ Max Flow (Row 19)	6.8 | 3.31 | 4.3	kgf/cm^2-g	Direct	Mandatory	Extract inlet pressure from '@ Max Flow' column of Row 19.	6.8 kgf/cm^2-g | 3.31 kgf/cm^2-g | 4.3 kgf/cm^2-g	No	No	P1 Max; Upstream Pressure Max; Inlet P Max	bar-g; kPa-g; psi-g; kgf/cm^2-g; kgf/cm^2-a	Row 12	Inlet Press.	Check gauge vs absolute units. Enquiry DS uses kgf/cm^2-g (gauge). If absolute ('a'), subtract 1.033 to convert to gauge.	
15	**Inlet Pressure (NOR)**	Inlet Pressure @ Norm Flow (Row 19)	6.8 | 3.31 | 4.33	kgf/cm^2-g	Direct	Mandatory	Extract inlet pressure from '@ Norm. Flow' column of Row 19.	6.8 kgf/cm^2-g | 3.31 kgf/cm^2-g | 4.33 kgf/cm^2-g	No	No	P1 Norm; Upstream Pressure Normal; Inlet P Normal	Same as Max	Row 12	Inlet Press.		
16	**Inlet Pressure (MIN)**	Inlet Pressure @ Min Flow (Row 19)	6.9 | 3.32 | 4.4	kgf/cm^2-g	Direct	Mandatory	Extract inlet pressure from '@ Min. Flow' column of Row 19.	6.9 kgf/cm^2-g | 3.32 kgf/cm^2-g | 4.4 kgf/cm^2-g	No	No	P1 Min; Upstream Pressure Min	Same as Max	Row 12	Inlet Press.		
17	**Outlet Pressure (MAX)**	Not directly stated — derived from Rows 19 & 20	Calculated: 6.8 - 6.57 = 0.23 | 3.31 - 0.3 = 3.01 | 4.3 - 1.72 = 2.58	kgf/cm^2-g	Derived	Mandatory	Outlet Pressure = Inlet Pressure (Row 19) − Pressure Drop (Row 20) for each case. Sv2 requires explicit outlet pressure.	0.23 kgf/cm^2-g | 3.01 kgf/cm^2-g | 2.58 kgf/cm^2-g	No	No	P2 Max; Downstream Pressure Max; Back Pressure Max	—	Row 13	Outlet Press.	If result is negative, check whether Enquiry DS uses absolute rather than gauge pressure. Flag for engineer confirmation.	
18	**Outlet Pressure (NOR)**	Not directly stated — derived from Rows 19 & 20	Calculated: 6.8 - 6.6 = 0.20 | 3.31 - 0.31 = 3.00 | 4.33 - 1.74 = 2.59	kgf/cm^2-g	Derived	Mandatory	Outlet Pressure NOR = Inlet Press NOR (Row 19 NOR) − Pressure Drop NOR (Row 20 NOR)	0.20 kgf/cm^2-g | 3.00 kgf/cm^2-g | 2.59 kgf/cm^2-g	No	No	P2 Norm; Downstream Pressure Normal	—	Row 13	Outlet Press.		
19	**Outlet Pressure (MIN)**	Not directly stated — derived from Rows 19 & 20	Calculated: 6.9 - 6.7 = 0.20 | 3.32 - 0.32 = 3.00 | 4.4 - 1.83 = 2.57	kgf/cm^2-g	Derived	Mandatory	Outlet Pressure MIN = Inlet Press MIN (Row 19 MIN) − Pressure Drop MIN (Row 20 MIN)	0.20 kgf/cm^2-g | 3.00 kgf/cm^2-g | 2.57 kgf/cm^2-g	No	No	P2 Min; Downstream Pressure Min	—	Row 13	Outlet Press.		
20	**Pressure Drop (MAX)**	Pressure Drop @ Max Flow (Row 20)	6.57 | 0.3 | 1.72	kgf/cm^2	Direct	Mandatory	Extract pressure drop from '@ Max Flow' column of Row 20.	6.57 kgf/cm^2 | 0.3 kgf/cm^2 | 1.72 kgf/cm^2	No	No	dP Max; Delta P Max; Differential Pressure Max; ΔP	bar; kPa; psi; kgf/cm^2	Row 14	Pressure drop	Pressure drop at max flow gives the largest Cv but may not be the most critical case for control.	
21	**Pressure Drop (NOR)**	Pressure Drop @ Norm Flow (Row 20)	6.6 | 0.31 | 1.74	kgf/cm^2	Direct	Mandatory	Extract pressure drop from '@ Norm. Flow' column of Row 20.	6.6 kgf/cm^2 | 0.31 kgf/cm^2 | 1.74 kgf/cm^2	No	No	dP Norm; Delta P Normal	Same as Max	Row 14	Pressure drop		
22	**Pressure Drop (MIN)**	Pressure Drop @ Min Flow (Row 20)	6.7 | 0.32 | 1.83	kgf/cm^2	Direct	Mandatory	Extract pressure drop from '@ Min. Flow' column of Row 20.	6.7 kgf/cm^2 | 0.32 kgf/cm^2 | 1.83 kgf/cm^2	No	No	dP Min; Delta P Min	Same as Max	Row 14	Pressure drop		
23	**Temperature (MAX)**	Inlet Temperature @ Max Flow (Row 21)	99.9 | 160 | 40	°C	Direct	Mandatory	Extract temperature from '@ Max Flow' column of Row 21.	99.9 °C | 160 °C | 40 °C	Yes	No	T Max; Process Temperature Max; Fluid Temp Max	°C; °F; K	Row 15	Temperature	High temperature affects packing selection, bonnet type, and body material grade.	
24	**Temperature (NOR)**	Inlet Temperature @ Norm Flow (Row 21)	99.9 | 160 | 40	°C	Direct	Mandatory	Extract temperature from '@ Norm. Flow' column of Row 21.	99.9 °C | 160 °C | 40 °C	Yes	No	T Norm; Process Temperature Normal	Same as Max	Row 15	Temperature		

Serial Number	Sv2 Field Name	Enquiry Field Name (Customer Spec)	Enquiry Field Value (Customer Spec)	Units	Mapping Category	Field Requirement	Logic	Value for Sv2 Field	Service Dependency	Sector Dependency	Similar Field Terminologies	Similar Field Values	SV2 Spec Row Number	Sv2 Spec Field Name	Notes and Validations	Additional Information
25	**Temperature (MIN)**	Inlet Temperature @ Min Flow (Row 21)	99.99 | 160 | 40	°C	**Direct**	**Mandatory**	Extract temperature from '@ Min. Flow' column of Row 21.	99.99 °C | 160 °C | 40 °C	Yes	No	T Min; Process Temperature Min	Same as Max	Row 15	Temperature		
26	**Specific Gravity / Molecular Weight**	Inlet Density / Specific Gravity / Molecular Mass (Row 22)	955 kg/m^3 (SG=0.955) | — (steam) | 28.96 (MW)	kg/m^3 or dimensionless or g/mol	**Derived**	**Mandatory**	Liquid service: Row 22 gives density in kg/m^3 → divide by 1000 to get SG (e.g. 955/1000 = 0.955). Gas service: Row 22 gives MW in g/mol (e.g. 28.96 for air). Steam service: leave blank; Sv2 uses steam tables.	SG = 0.955 | Blank (steam) | MW = 28.96 g/mol	Yes	No	Spec. Gravity; SG; Specific Gravity; Molecular Weight; MW; Density; Relative Density	kg/m^3; kg/L; g/cm^3; dimensionless SG; g/mol	Row 17	Spec. Gravity	Units must be verified. If Row 22 header says 'kg/m^3' for a liquid, divide by 1000 for SG. If already dimensionless, use directly.	
27	**Viscosity**	Inlet Viscosity (Row 24)	0.276 | 0.014 | 0.0191	cP	**Direct**	**Mandatory**	Copy viscosity value from Row 24 of Enquiry DS.	0.276 cP | 0.014 cP | 0.0191 cP	Yes	No	Viscosity; Dynamic Viscosity; Kinematic Viscosity; µ; Centipoise	cP; mPa·s; cSt (note: cSt ≠ cP for density ≠ 1)	Row 18	Viscosity	Viscosity affects Fp correction factor for high-viscosity liquids. If in cSt, convert: cP = cSt × SG.	
28	**Compressibility Factor (Z)**	Inlet Compressibility Factor (Row 23)	— (liquid; not applicable) | — (steam) | 0.999	Dimensionless	**Direct**	**Conditional**	Extract Z from Row 23. Applicable for gas/vapour only. Leave blank for liquid and steam (Sv2 applies steam tables).	N/A (liquid) | N/A (steam) | 0.999	Yes	No	Z Factor; Gas Compressibility; Compressibility; Z	0.95 to 1.05 typical range	Row 16	Z	For ideal gases Z = 1.0. If Row 23 is blank for a gas service, use Z = 1.0 as default.	
29	**Specific Heats Ratio (K)**	Inlet Specific Heats Ratio (Row 25)	— (liquid; not applicable) | 1.317 (steam) | 1.4 (air)	Dimensionless	**Direct**	**Conditional**	Extract K from Row 25. Applicable for gas and steam only. Leave blank for liquid services.	N/A (liquid) | 1.317 | 1.4	Yes	No	Isentropic Exponent; Cp/Cv; Gamma; K; Specific Heat Ratio	Air = 1.4; Steam ≈ 1.3; Nitrogen = 1.4; CO2 = 1.3	Row 18	K	K affects the choked flow (critical pressure drop ratio) calculation. If not stated for a gas service, use 1.4 as conservative default for diatomic gases.	
30	**Vapour Pressure**	Inlet Vapour Pressure (Row 26)	1.031 kgf/cm^2-a | — (steam; not applicable) | — (gas; not applicable)	kgf/cm^2-a	**Direct**	**Conditional**	Extract vapour pressure from Row 26. Applicable for liquid services only. Used to check cavitation and flashing.	1.031 kgf/cm^2-a | N/A | N/A	Yes	No	Pv; Vapour Pressure; Psat; Liquid Vapour Pressure	kgf/cm^2-a; bar-a; psia; kPa-a	Row 16	Vapour Press.	Must be in absolute pressure units. If given as gauge, add 1.033 kgf/cm^2 to convert to absolute.	
31	**Valve Type**	Body Type (Row 32)	Single Seat Globe | Single Seat Globe | Single Seat Globe	—	**Direct**	**Mandatory**	Copy body type from Row 32 of Enquiry DS verbatim.	Single Seat Globe | Single Seat Globe | Single Seat Globe	Yes	No	Body Type; Valve Type; Valve Body Style; Body Style	Single Seated Globe; Double Seated Globe; Butterfly; Ball; Angle; Y-Pattern	Row 27	Type	Valve type is always stated by the customer in this format. Do not leave for engineer to determine.	
32	**Body Material**	Body Material (Row 36)	F.S. ASTM A105 Normalized | C.S. ASTM A216 WCB | C.S. ASTM A216 WCB	—	**Direct**	**Mandatory**	Copy body material specification verbatim from Row 36 of Enquiry DS. This is fully stated by the customer.	F.S. ASTM A105 Normalized | ASTM A216 Gr. WCC | ASTM A216 Gr. WCC	Yes	Yes	Body Material; Body MOC; Body Grade; Pressure Boundary Material	A105; A216 WCB; A216 WCC; A351 CF8M; A351 CF3M; A182 F316	Row 29	Body Material	Note: Enquiry DS says 'A216 WCB' while KSB MIL DS shows 'A216 Gr. WCC' — verify with engineer. WCB and WCC are different grades. Do not silently change.	
33	**Bonnet Type**	Bonnet Type / Material (Row 37)	Bolted Flange A105 | Bolted Flange A216 WCB | Bolted Flange A216 WCB	—	**Direct**	**Mandatory**	Copy bonnet type and material from Row 37 of Enquiry DS.	Bolted Flange A105 | Bolted Flange A216 WCB | Bolted Flange A216 WCB	Yes	No	Bonnet Type; Bonnet; Bonnet Style; Bonnet Material	Standard; Extended; Finned Extension; Cryogenic; Bolted Flange; Pressure Seal	Row 35	Bonnet Type	Extended or finned bonnet required for steam above 200°C or cryogenic services. Check pipe insulation (Row 13) for guidance.	
34	**Flow Direction**	Flow Direction (Row 38)	FTC (Flow-To-Close) | FTO (Flow-To-Open) | FTO (Flow-To-Open)	—	**Direct**	**Mandatory**	Copy flow direction from Row 38 of Enquiry DS. FTC = Flow-To-Close; FTO = Flow-To-Open.	Flow to close | Flow to open | Flow to open	Yes	No	Flow Direction; FTC; FTO; Flow-To-Close; Flow-To-Open; Flow Under Seat	FTC; FTO; Over the seat; Under the seat	Row 31	Flow Direction	Flow direction affects actuator unbalance forces and determines stability. Customer-specified; do not change.	
35	**End Connection & Rating**	End Connec. & Rating (Row 35)	1.5" 300# RF | 8" 300# RF | 4" 300# RF	—	**Direct**	**Mandatory**	Copy end connection type, size, and rating from Row 35 of Enquiry DS.	Flanged-ASME 300 RF | Flanged-ASME 300 RF | Flanged-ASME 300 RF	No	No	End Connection; Flange Rating; ANSI Class; Flange Type; Body Rating	ASME 150 RF; ASME 300 RF; ASME 600 RF; ASME 900 RTJ; Butt Weld	Rows 30/32/33	End Conn. (Inlet/Outlet)	Body size shown in Row 33 ('By Vendor') means KSB MIL selects from catalogue after Cv calculation. Rating determines the body design pressure.	
36	**Flow Characteristic**	Characteristic (Rated Cv / Characteristic Row 34)	Linear | Equal% | Linear	—	**Derived**	**Mandatory**	Row 34 shows 'By Vendor | Equal% | By Vendor'. For 'By Vendor', engineer selects based on process control requirements: Equal Percentage for pressure-drop ratio > 0.3; Linear for most other cases.	Linear | Equal % | Linear	Yes	No	Characteristic; Flow Characteristic; Inherent Characteristic; Cv Characteristic	Linear; Equal Percentage; Quick Opening; Modified Parabolic	Row 40	Characteristic	Equal Percentage selected when the valve pressure drop varies significantly across the flow range. Linear when pressure drop is relatively constant.	
37	**Trim / Plug Type**	Trim Type (Row 42)	Low Noise Trim | Contoured Trim | Low Noise Trim	—	**Direct**	**Mandatory**	Copy trim type from Row 42 of Enquiry DS verbatim.	Low Noise Trim (Anti-Cavitation) | Contoured Trim | Low Noise Trim	Yes	No	Trim Type; Plug Type; Trim Style; Plug Style; Cage Type	Standard Contoured; Low Noise; Anti-Cavitation; Parabolic; V-Port; Characterized	Row 42	Plug Type	Low Noise trim is specified because SPL at max flow approaches or exceeds allowable. Verify SPL result after Cv calculation.	
38	**Plug Material**	Plug / Ball / Disk Material (Row 44)	410 (13Cr) | 410 (13Cr) | 410 (13Cr)	—	**Direct**	**Mandatory**	Copy plug material from Row 44 of Enquiry DS verbatim.	SS 410 + Stellite No.6 | SS 410 + Stellite No.6 | SS 410 + Stellite No.6	Yes	Yes	Plug Material; Ball Material; Disk Material; Trim Material; Plug MOC	SS 316; SS 410; 17-4PH; Stellite overlay; Hastelloy C	Row 46	Plug Material	Enquiry DS states '410 (13Cr)'; KSB MIL DS output shows 'SS 410 + Stellite No.6' — the Stellite hard-facing is a KSB MIL standard addition for corrosion and erosion resistance.	
39	**Seat Material**	Seat Material (Row 45)	St Gr6 (CoCrAlloy) | St Gr6 (CoCrAlloy) | St Gr6 (CoCrAlloy)	—	**Direct**	**Mandatory**	Copy seat material from Row 45 of Enquiry DS verbatim.	SS 410 + Stellite No.6 | SS 410 + Stellite No.6 | SS 410 + Stellite No.6	Yes	Yes	Seat Material; Seat Ring Material; Seat MOC	SS 316; SS 410; Stellite; Monel; Inconel	Row 47	Seat Material	'St Gr6' = Stellite Grade 6 (Cobalt-Chromium alloy). KSB MIL DS shows seat as SS 410 + Stellite overlay. Confirm mapping with engineer.	
40	**Seat Leakage Class**	Tightness Requirements (Row 7)	ANSI IV (standard) | ANSI IV (standard) | ANSI V	—	**Derived**	**Mandatory**	Map customer tightness requirement to FCI 70.2 leakage class: ANSI IV → IV (FCI 70.2); ANSI V → V (FCI 70.2); ANSI VI → VI (FCI 70.2)	IV (FCI 70.2) | IV (FCI 70.2) | V (FCI 70.2)	No	No	Leakage Class; Seat Leakage; Tightness Class; Shut-off Class	Class I; Class II; Class III; Class IV; Class V; Class VI; Zero Leakage	Row 41	Seat Leakage Class	Class V required for Tag 3 (ANSI V). Class V requires lapped metal seat. Increases cost and lead time. Flag for review.	
41	**Guiding**	Guiding / No. of Ports (Row 41)	By Vendor / 1 (all tags)	—	**Engineered**	**Optional**	'By Vendor' means KSB MIL selects guiding arrangement. Standard for single-seated globe: Heavy Top Guiding for Tag 1 and 3; Cage Guiding for Tag 2 (heavy-duty steam).	Heavy Top Guiding | Cage Guiding | Heavy Top Guiding	Yes	No	Guiding; Guide Type; Plug Guiding; Top-and-Bottom Guiding	Top Guided; Cage Guided; Top-and-Bottom Guided; Port Guided	Row 45	Guiding	Guiding selection affects valve stability at high pressure drops. KSB MIL selects based on valve series and dP ratio.	
42	**Actuator Type**	Type (Actuator) (Row 50)	Spring Diaphragm | Spring Diaphragm | Spring Diaphragm	—	**Direct**	**Mandatory**	Copy actuator type from Row 50 of Enquiry DS.	Pneumatic Spring Diaphragm | Pneumatic Spring Diaphragm | Pneumatic Spring Diaphragm	No	No	Actuator Type; Actuator Style; Pneumatic Actuator; Diaphragm Actuator	Spring Diaphragm; Piston; Scotch-Yoke; Electric; Electro-Hydraulic	Row 51	Type	All three tags use Spring Diaphragm type — the most common pneumatic actuator. Piston actuator used for high shut-off or large bore valves.	
43	**Actuator Action / Air Failure**	Power Failure Position (Row 9) + Air Failure Valve (Row 52)	Open (R9) / Open (R52) | Close (R9) / Close (R52) | Open (R9) / Open (R52)	—	**Derived**	**Mandatory**	Row 9 = Power Failure Position (safe position on air loss). Map: Fail Open → Air to Close (ATC); Fail Close → Air to Open (ATO). Row 52 confirms. Cross-check both rows.	Air to Close (ATC) | Air to Open (ATO) | Air to Close (ATC)	Yes	No	Fail Position; Fail Safe; Air Failure; FO; FC; Air to Close; Air to Open; ATC; ATO	FO (Fail Open); FC (Fail Close); FL (Fail Last); Air to Close; Air to Open	Rows 53/54	Action / Air Failure	Actuator colour coding follows: ATC (Fail Open) → GREEN actuator; ATO (Fail Close) → RED actuator. See KSB MIL DS Sheet 2 Row 112.	
44	**Supply Pressure**	Available Air Supply Pressure (Row 8)	Min 3.5 kgf/cm^2 / Max 7 kgf/cm^2 (all tags)	kgf/cm^2	**Derived**	**Mandatory**	Use minimum available supply pressure from Row 8. Convert to psi: × 14.223. Round to nearest standard value. 3.5 kgf/cm^2 × 14.223 ≈ 50 psi.	50 psi (all tags)	No	No	Air Supply; Instrument Air; IA Supply; Air Supply Pressure; Pneumatic Supply	35 psi; 50 psi; 60 psi; 80 psi	Row 56	Supply Press.	Use minimum supply to ensure actuator can stroke even at lowest supply condition. Bench range must be set below supply pressure minus positioner overhead (~5 psi).	
45	**Bench Range**	Bench Range (Row 54)	By Vendor | By Vendor | By Vendor	psi	**Engineered**	**Mandatory**	'By Vendor' on all three tags → KSB MIL calculates bench range from actuator sizing (spring rate × travel + preload). Typical values: 3-15 psi, 6-30 psi. KSB MIL DS shows 3-15 psi (Tag1) and 6-30 psi (Tag2).	3-15 psi (Tag1) | 6-30 psi (Tag2) | 5-18 psi (Tag3)	No	No	Bench Range; Spring Range; Spring Calibration; Air Range; Bench Set	3-15 psi; 6-30 psi; 3-27 psi; 6-18 psi	Row 55	Bench Range	Bench range calculated by actuator sizing to ensure adequate shut-off thrust. Cannot be determined from Enquiry DS alone — leave for engineer.	
46	**Shut-off Pressure**	Actuator Design Pressure (Row 10)	Not stated in Enquiry DS — derived from Row 27 Design Pressure	kgf/cm^2	**Derived**	**Mandatory**	Shut-off pressure = maximum design pressure of valve (Row 27 MAX). KSB MIL DS Row 57 shows this value.	12 kgf/cm^2 (Tag1) | 8 kgf/cm^2 (Tag2) | 10 kgf/cm^2 (Tag3)	No	No	Shut-off Pressure; Actuator Rating; MAOP; Max Operating Pressure; Differential Shut-off Pressure	—	Row 57	Shut off Press.	Used to size actuator spring and diaphragm area to ensure valve closes tightly against max design pressure.	
47	**Handwheel**	Handwheel (Row 53)	— (dash; not specified) | — (dash) | Side Mounted	—	**Direct**	**Optional**	Copy handwheel requirement from Row 53. '—' or blank = No handwheel. 'Side Mounted' = side-mounted handwheel assembly.	No/NA | No/NA | Yes/6A2 (Side Mounted)	No	No	Handwheel; Manual Override; Manual Operator; Side Handwheel; Top Handwheel	No; NA; Side Mounted; Top Mounted; Integral	Row 58	Hand Wheel/Model	Tag 3 (Plant Air) requires side-mounted handwheel. This is a manual override device — allows operator to position valve when air supply is unavailable.	
48	**Positioner Type**	Not stated in Enquiry DS (Row 56 = VTA/VTA)	VTA / VTA (all tags)	—	**Engineered**	**Mandatory**	VTA = Vendor To Advise. KSB MIL selects standard smart positioner: Smart Single Acting (HART). Selection standard for all modulating control valves.	Smart Single Acting	No	No	Positioner Type; Positioner Style; Electro-Pneumatic Positioner	Single Acting; Double Acting; Smart; Conventional; Digital	Row 72	Type	Smart positioner integrates position transmitter — eliminates need for separate position transmitter hardware.	
49	**Positioner Make**	Positioner MFR (Row 56)	VTA (all tags)	—	**Derived**	**Mandatory**	VTA → KSB MIL standard selection: METSO. If customer specifies a make (not VTA), copy directly.	METSO	No	No	Positioner Manufacturer; Positioner Make; Brand	METSO; SAMSON; FISHER; ABB; SIEMENS; ROTORK	Row 73	Make	METSO ND9103 is the KSB MIL standard positioner. Alternative approved make: SAMSON 3730.	

Serial Number	Sv2 Field Name	Enquiry Field Name (Customer Spec)	Enquiry Field Value (Customer Spec)	Units	Mapping Category	Field Requirement	Logic	Value for Sv2 Field	Service Dependency	Sector Dependency	Similar Field Terminologies	Similar Field Values	SV2 Spec Row Number	Sv2 Spec Field Name	Notes and Validations	Additional Information
50	**Positioner Model**	Positioner Model (Row 56)	VTA (all tags)	—	**Derived**	**Mandatory**	VTA → KSB MIL standard model: ND9103-HX-T (HART communication, aluminium housing, ATEX certified).	ND9103-HX-T	No	No	Positioner Model; Model Number	ND9103-HX-T; TZID-C; 3731-3; DVC6200	Row 74	Model	ND9103-HX-T: HX = HART with XP rating; T = with position transmitter integrated.	
51	**Positioner Protocol / Casing**	Signal Inlet (Row 57)	4-20mA+HART / 3-15Psi (all tags)	—	**Derived**	**Mandatory**	Signal '4-20mA+HART' → Protocol = HART. Standard casing = Aluminium (IP66). Combined field: HART/Aluminium.	HART/Aluminium	No	No	Communication Protocol; Positioner Protocol; Housing Material; Casing	HART; FOUNDATION Fieldbus; PROFIBUS PA; 4-20mA only	Row 75	Protocol / casing		
52	**Positioner Action / Input Signal**	Signal Inlet (Row 57) + Increase Signal Valve (Row 58)	4-20mA+HART, Increase Signal = Close (Tags 1&3) / Open (Tag 2)	mA	**Derived**	**Mandatory**	Combine action sense and signal: 'Increase Signal → Close' = Direct Acting. 'Increase Signal → Open' = Reverse Acting. All tags use 4-20 mA input. Result: 'Direct/4-20 mA' for ATC valves.	Direct/4-20 mA (all tags)	Yes	No	Action; Input Signal; Positioner Action; Direct Acting; Reverse Acting	Direct/4-20mA; Reverse/4-20mA; Direct/3-15psi; Split-range	Row 76	Action/Input Signal	'Increase Signal → Close' means as signal increases, valve closes. This is Direct Acting positioner (increasing air closes valve = ATC). For ATO valves, Increase Signal → Open is	
53	**Positioner Certification**	Area Classification (Row 4)	Zone 1 Gr. IIC T3 (all tags)	—	**Derived**	**Mandatory**	Zone 1 IIC T3 → Intrinsic Safety certification required: 'Intrinsic Safe Ex ia IIC, T6, with Weather Proof IP 66 Enclosure'	Intrinsic Safe Ex ia IIC, T6, WP IP66	No	Yes	ATEX Certification; Hazardous Area Certificate; Ex Certification; IS Certificate	Ex ia; Ex d; EEx e; Ex n; PESO Certificate	Row 77	Certification	Zone 1 requires Intrinsic Safety (Ex ia) or Flameproof (Ex d) certification. KSB MIL standard uses Ex ia for positioner. PESO certificate also required (see quality requirements).	
54	**Position Transmitter**	Position Transmitter (Row 61)	Yes. 4-20mA+Hart (all tags)	—	**Derived**	**Conditional**	'Yes' → Position transmitter is integral to the smart positioner (METSO ND9103). No separate device. Mounting: Integral to Smart Positioner. Output: 24 VDC / 4-20 mA (voltage from Row 62).	Integral to Smart Positioner / 24 VDC / 4-20mA	No	No	Position Transmitter; Stem Position Transmitter; Position Feedback; LVDT	Integral; Separate; Remote Mount; LVDT; Hall Effect	Rows 86–89	Type / Make/Model / Supply/Output	If Row 61 = 'No', populate Rows 86–92 as 'Not Required'. If a separate transmitter is specified (non-integral), treat as Engineered selection.	
55	**Airset Make/Model/Qty**	Air Set MFR / Model (Row 72)	VTA / VTA / VTA (all tags)	—	**Derived**	**Mandatory**	VTA → KSB MIL standard: Shavo SB77-276-M6KN-N, 1 No. Gauges: Yes (Row 74) → include pressure gauge.	Shavo/SB77-276-M6KN-N/1 No	No	No	Air Set; Filter Regulator; AFR; Airset; Filter Reg Unit	Shavo; Norgren; SMC; Parker; Watts	Row 67	Make/Model/Qty	Airset conditions instrument air (filters and regulates) before it enters positioner and actuator. Filter: 5 micron standard.	
56	**Airset Set Pressure / Filter**	Airset Set Pressure (Row 73) / Filter (Row 74)	By Vendor / Gauge: Yes (all tags)	—	**Derived**	**Mandatory**	Set pressure = minimum available air supply (Row 8 MIN) = 3.5 kgf/cm² ≈ 50 psi. Filter size: standard = 5 micron. Gauge: Yes.	Set Pressure: By Vendor / Filter: 5 Micron / Gauge: Yes	No	No	Set Pressure; Filter Size; Micron Rating; Filter Element	5 micron; 25 micron; 40 micron; 0.3 micron	Rows 68/73/74	Filter Size / Set Press / Gauge		
57	**Painting Scheme**	Not in standard Enquiry DS body — from PO/project	Not stated in Enquiry DS	—	**Engineered**	**Mandatory**	KSB MIL standard: Body — SPECIAL | Actuator — SPECIAL. Painting procedure reference from project PO (e.g. CP 121_PK233545).	Body - SPECIAL | Actuator - SPECIAL	No	Yes	Painting; Paint Scheme; Surface Treatment; Colour Code	RAL 7001; IT 228HS; Epoxy; Primer + Top coat	Row 111	Painting Scheme	Painting scheme varies by project. Always refer to PO painting specification. Body colour: Grey (IT 228HS). Actuator colour: GREEN (fail-open) or RED (fail-close).	
58	**Actuator Colour**	Power Failure Position (Row 9)	Open | Close | Open	—	**Derived**	**Mandatory**	Fail Open (Row 9 = Open) → Actuator colour GREEN. Fail Close (Row 9 = Close) → Actuator colour RED. Body always GREY (IT 228HS).	GREEN | RED | GREEN	No	No	Actuator Colour; Actuator Color; Body Colour; Colour Code	Green; Red; Yellow; Grey	Row 112	Body Colour / Actuator Colour	Colour coding is a visual safety indicator of fail-safe position. Standard: Green = Fail Open; Red = Fail Close.	
59	**Cable Gland Type**	Not in Enquiry DS	Not stated	—	**Engineered**	**Mandatory**	KSB MIL standard: Double compression type. Certification derived from Area Classification (Row 4).	Double compression type	No	No	Cable Gland; Gland Type; Cable Entry Device	Single compression; Double compression; Stopping plug; EMC gland	Row 130	Type		
60	**Cable Gland Certification**	Area Classification (Row 4)	Zone 1 Gr. IIC T3 (all tags)	—	**Derived**	**Mandatory**	Zone 1 IIC → Explosion Proof: Ex d IIA, IIB, IIC. Material: SS 304 + PVC Hood. Cable Type: Flexible.	Explosion Proof Ex d IIA, IIB, IIC	No	Yes	Cable Gland Certification; Gland Cert; Hazardous Area Gland; ATEX Gland	Ex d; Ex e; Ex n	Row 131	Certification	Cable gland certification must be independent of positioner certification. Even if positioner is Ex ia, cable gland must be rated Ex d for Zone 1.	
61	**IBR Applicability**	Process Fluid (Row 14/15) + Specification Notes (DS Page 2)	Not steam (Tag1/3); IBR Applicable (Tag2 — process notes state 'IBR Applicable')	—	**Derived**	**Mandatory**	If fluid = steam AND 'IBR' appears in DS Page 2 Spec/Process Notes → IBR applies. Valve must be tested and certified by IBR per Appendix L. Otherwise 'IBR Not Required'.	IBR Not Required (Tag1/3) | IBR Certificate in Form IIIC (Tag2)	Yes	Yes	IBR; Indian Boiler Regulation; Boiler Certificate; IBR Certificate	IBR Applicable; IBR Required; Form IIIC; Not applicable to IBR	Rows 137/145	Special Testing/Quality Requirements	IBR certificate requires third-party inspection during manufacturing. Increases lead time by 4-6 weeks. Flag immediately when identified.	
62	**NDT — RT Extent**	Not in Enquiry DS directly — standard requirement	Standard for all castings (Body, Bonnet, Flanges)	—	**Derived**	**Mandatory**	Standard KSB MIL requirement for all projects: 10% RT per lot for Body/Bonnet/Com.Flange/Expander/LNP. Trigger: always apply unless customer explicitly waives.	[0510] RT extent 10% per lot — BODY/BONNET/COM.FLANGE/EXP/LNP	No	No	RT; Radiographic Testing; X-Ray; NDE; NDT	10% RT; 100% RT; Spot RT; UT (Ultrasonic)	Row 138	Special Testing/Quality Requirements		
63	**NDT — PMI Test**	Not in Enquiry DS directly — standard requirement	Standard for all trim parts	—	**Derived**	**Mandatory**	Standard KSB MIL requirement: PMI Test (Standard grade) for all trim parts. Trigger: always apply.	[1801] PMI Test (Std) applicable for TRIM PARTS	No	No	PMI; Positive Material Identification; Material Test; XRF; Chemical Analysis	PMI Standard; PMI Full; 100% PMI; PMI on castings	Row 142	Special Testing/Quality Requirements	PMI verifies that trim materials match specification (e.g. confirms SS 410 is not SS 304 supplied in error).	
64	**NDT — IGC Test**	Trim/Seat Material (Rows 44/45) — triggered by Austenitic SS	Plug: 410 (13Cr) — not austenitic, trigger NOT activated for these tags	—	**Derived**	**Conditional**	IGC test triggered when plug/seat material contains Austenitic Stainless Steel (SS316, CF8M, CF3M, 304 etc.). SS 410 (martensitic) and Stellite do NOT trigger IGC. For these three tags: IGC row is	[1707] IGC testing as per Prac.E, 250× microstructure (applicable for Aust.SS Trim Parts)	Yes	No	IGC; Intergranular Corrosion; Sensitisation Test; ASTM A262 Prac E	IGC Test; Weld Decay Test; Sensitisation	Row 139	Special Testing/Quality Requirements	If trim material is changed to SS316 or CF8M in a future revision, IGC test becomes mandatory.	
65	**Valve Operating Signature**	Not in Enquiry DS — standard requirement	Standard for all control valves	—	**Derived**	**Mandatory**	Standard KSB MIL requirement: valve operating signature (stem position vs control signal) to be supplied on CD/soft copy.	[2805] Valves operating signature in the form of CD	No	No	Operating Signature; Valve Signature; Stroke Test Record; Loop Test	CD; PDF; Digital Report; FAT Record	Row 143	Special Testing/Quality Requirements		
66	**PESO Certificate**	Area Classification (Row 4)	Zone 1 Gr. IIC T3 (all tags)	—	**Derived**	**Mandatory**	Hazardous area (Zone 1 or Zone 2) → PESO certificate required for all electrical accessories (positioner, solenoid valve if any, limit switches if any).	[2301] PESO Certificate for All Electrical Accessories	No	Yes	PESO; CCOE; Hazardous Area Certificate; Electrical Accessory Certificate	PESO; CCOE; Statutory approval	Row 144	Special Testing/Quality Requirements	PESO (Petroleum and Explosives Safety Organisation) is the Indian statutory body. Certificate must be obtained before despatch. Add 4-6 weeks to delivery if not pre-certified.	